BUSINESS DOCUMENT MANAGEMENT SYSTEM

Software Requirements & Technical Handoff Document

Project Type: Web-based Business Document Management System
Version: 1.0
Frontend & Backend: Next.js + TypeScript
Database: PostgreSQL
ORM: Prisma
Architecture: Modular Monolithic Architecture
Initial Platform: Web Admin Dashboard
Future Platform: Android & iOS Application

---

1. PROJECT OVERVIEW

Develop a web-based Business Document Management System to digitalize the manual document generation process of a business owner who manages multiple businesses, government tender projects, and local projects.

The system will replace manually prepared printed documents with a centralized digital platform.

Users should be able to:

- Manage multiple business profiles.
- Manage customers, suppliers, and contractors.
- Manage government tender and local projects.
- Maintain reusable product/item information.
- Generate different types of business documents.
- Automatically calculate document totals.
- Generate unique document numbers.
- Preview, print, and download documents as PDF.
- Maintain document history and reports.

The system must support two existing businesses under the same owner.

2. BUSINESS PROFILES

Initially, the system will support the following businesses:

Business 01: Apurbo Trade International (ATI)

- Business Name: অপূর্ব ট্রেড ইন্টারন্যাশনাল
- Business Type: Trading / Construction Materials Supply
- Contact Numbers: 01680-966570, 01627-631140
- Business Address: মীরপুর/মির্জাপুর রোড, সিলকোট, ঢাকা-১২২৯ (পুরোনো ফর্মে প্রদর্শিত ঠিকানা; ব্যবহারের আগে যাচাই করতে হবে)

Business 02: ARJ Tex Accessories

- Business Type: Garments Accessories Manufacturer, Importer & Supplier
- Office & Factory: House #43/2, Dohadia Molla Para, Dakshinkhan, Uttara, Dhaka-1230
- Contact Numbers: 01680966570, 01715 659 599
- Email: arjtexaccessories@gmail.com

Requirements:

- Both businesses must be manageable from a single admin account.
- Implement a Business Switcher in the dashboard.
- Each business must have separate logo, contact details, address, document numbering, templates, and settings.
- All documents must be associated with a specific business.
- Business information must be stored as a snapshot when generating a document to preserve historical accuracy.

3. DOCUMENT TYPES

The system must support eight document categories.

1. Invoice
2. Delivery Challan
3. Cash Memo
4. Money Receipt
5. Quotation
6. Purchase Order
7. Work Order
8. Other Documents

Each document type must have its own field configuration and print template.

All document types should use a unified document generation engine instead of separate CRUD implementations.

4. EXISTING DOCUMENT ANALYSIS

The following information was extracted from the provided document samples.

4.1 ATI Cash Memo / Form

Fields:

- Business Logo
- Business Name
- Contact Numbers
- Document Number
- Date
- Customer Name
- Customer Address
- Product Description
- Vehicle Number
- Amount
- Amount in Words
- Seller Signature

The uploaded blank form also contains two identical copies side by side. The final print layout must support configurable multiple copies per page.

Note: The uploaded form is labelled as a Cash Memo/Challan-style form. Its exact operational use must be confirmed with the client.

4.2 ARJ Tex Delivery Challan

Fields:

- Business Logo and Information
- Challan Number
- Date
- Customer Name
- Customer Address
- Mobile Number

- Product Description
- Quantity
- Unit
- Remarks
- Total Quantity
- Receiver Signature
- Authorized Signature

Sample data:

- Customer: Staff Fashion
- Address: Charagali, Gazipur
- Item: Gross Grant Tape / Belt / 3.8mm / Offwhite
- Quantity: 450 Yards
- Item: Gross Grant Tape / Belt / 3.8mm / Mocha
- Quantity: 200 Yards
- Total Quantity: 650 Yards

Important: Delivery Challan does not necessarily require Rate or Amount fields.

4.3 ARJ Tex Invoice

Fields:

- Business Logo and Information
- Invoice Number
- Date
- Customer Name
- Customer Address
- Product Description
- Quantity
- Unit
- Rate
- Amount
- Subtotal
- Discount (optional)
- Tax (optional)
- Total
- Amount in Words
- Receiver Signature
- Authorized Signature

Sample:

- Invoice Number: 182

- Date: 10/05/2026
- Customer: Stuff Limited
- Address: Charagali, Tongi, Gazipur
- Product: Belt
- Variations: Summer Cherry, Mocha, Offwhite
- Quantity: 1800 Yards
- Rate: 17
- Total: BDT 30,600

The original handwritten document should be used as a reference for recreating the layout.

5. FUNCTIONAL MODULES

5.1 Dashboard

- Total documents
- Total invoices
- Total delivery challans
- Total projects
- Recent documents
- Recent activities
- Business-wise statistics
- Date-wise statistics

5.2 Business Management

- Create, update, and deactivate business profiles.
- Upload business logos.
- Manage address, phone, email, website.
- Configure document prefixes.
- Configure signatures.
- Manage business-specific templates.

5.3 Party Management

Manage all business-related parties in one module.

Party Types:

- Customer
- Supplier

- Contractor
- Both
- Other

Fields:

- Name
- Company Name
- Contact Person
- Phone
- Email
- Address
- Tax ID (optional)
- Notes

Each party must belong to a business.

5.4 Project Management

Support two project categories:

Government Tender

- Project Name
- Tender Number
- Tender ID
- Department / Authority
- Work Order Number
- Associated Customer / Authority
- Project Location
- Start Date
- End Date
- Description
- Status

Local Project

- Project Name
- Customer
- Location
- Description
- Start Date
- End Date
- Status

Project Types:

- TENDER
- LOCAL

A project can have multiple associated documents.

Project association must be optional when creating documents.

5.5 Product Management

Maintain a reusable product/item catalog.

Fields:

- Product Name
- Description
- SKU
- Unit
- Default Rate
- Business
- Active Status

Example:

- Belt — Yard — 17 BDT
- Cement — Bag
- Brick — Piece
- Sand — CFT

Users must also be able to add custom descriptions directly when generating documents without selecting a saved product.

Product inventory and stock tracking are outside the initial MVP scope.

5.6 Document Management

Features:

- Create document
- Edit draft document
- View document
- Duplicate document
- Issue document
- Cancel document

- Search documents
- Filter by business
- Filter by document type
- Filter by project
- Filter by customer
- Filter by date
- Print document
- Download PDF

Document Status:

- DRAFT
- ISSUED
- CANCELLED

Issued documents must not be directly edited or deleted. Cancellation must preserve the original document and its history.

5.7 Document Templates

Each business must support:

- Original Design
- Modern Design

Each document type can have its own template configuration.

Template settings:

- Logo visibility
- Header information
- Footer information
- Watermark
- Page size
- Page orientation
- Font
- Table columns
- Signature positions
- Number of copies per page

5.8 Reports

- Business-wise document report
- Project-wise document report

- Customer-wise document report
- Document type report
- Date-wise document report
- Issued / Draft / Cancelled report

---

6. DOCUMENT GENERATION WORKFLOW

1. Select Business.
2. Select Document Type.
3. Select Customer / Supplier.
4. Select Project (optional).
5. Enter document date and reference information.
6. Add document items.
7. Enter quantities, units, rates, and remarks.
8. Calculate totals automatically where applicable.
9. Enter additional information and terms.
10. Select Original or Modern template.
11. Preview document.
12. Save as Draft or Issue.
13. Generate document number upon issuing.
14. Print or download as PDF.

Additional functionality:

- Generate Invoice from an existing Delivery Challan.
- Duplicate an existing document.
- Reuse customer and item information.
- Allow modification of copied item quantities and rates before issuing.

---

7. DATABASE ARCHITECTURE

Database: PostgreSQL
ORM: Prisma

Use UUID as primary keys.

7.1 Main Database Tables

Table	Purpose
users	System users
business_members	User-business relationships
businesses	Business profiles
parties	Customers, suppliers, contractors
projects	Tender and local projects
products	Reusable product catalog
documents	All document records
document_items	Document line items
document_sequences	Automatic numbering
document_templates	Print template configurations
audit_logs	Activity history

Auth.js database adapter tables must be added if using its database-backed session strategy.

7.2 Relationships

- One User can access multiple Businesses.
- One Business can have multiple Users.
- One Business can have multiple Parties.
- One Business can have multiple Projects.
- One Business can have multiple Products.
- One Business can have multiple Documents.
- One Party can have multiple Projects.
- One Party can have multiple Documents.
- One Project can have multiple Documents.
- One Document can have multiple Document Items.
- One Product can be referenced by multiple Document Items.
- One Business can have multiple Document Templates.
- One Document can have multiple Audit Log entries.

7.3 Business Model

Fields:

- id
- name
- nameBn
- logo
- address
- phone
- email
- website
- footerText
- currency
- isActive
- createdAt
- updatedAt

7.4 Party Model

Fields:

- id
- businessId
- name
- companyName
- partyType
- contactPerson
- phone
- email
- address
- taxId
- notes
- isActive
- createdAt
- updatedAt

7.5 Project Model

Fields:

- id
- businessId
- partyId (nullable)
- name

- projectType
- tenderNumber (nullable)
- workOrderNumber (nullable)
- department (nullable)
- location
- description
- startDate
- endDate
- status
- createdAt
- updatedAt

7.6 Product Model

Fields:

- id
- businessId
- name
- description
- sku
- unit
- defaultRate
- isActive
- createdAt
- updatedAt

7.7 Document Model

Fields:

- id
- businessId
- partyId (nullable)
- projectId (nullable)
- templateId (nullable)
- createdById
- documentType
- documentNumber (nullable for draft)
- documentDate
- validUntil (nullable)
- referenceNumber
- subject

- partyRole
- businessSnapshot (JSONB)
- partySnapshot (JSONB)
- subtotal
- discountType
- discountValue
- discountAmount
- taxRate
- taxAmount
- totalAmount
- amountInWords
- notes
- terms
- customFields (JSONB)
- status
- issuedAt
- cancelledAt
- cancellationReason
- createdAt
- updatedAt

7.8 Document Item Model

Fields:

- id
- documentId
- productId (nullable)
- description
- quantity
- unit
- rate (nullable)
- amount (nullable)
- remarks
- sortOrder

Document item descriptions, units, and rates must be stored as snapshots. Later product catalog updates must not affect previously created documents.

7.9 Document Sequence Model

Fields:

- id
- businessId
- documentType
- fiscalYear
- prefix
- lastNumber
- padding

Unique constraint:
businessId + documentType + fiscalYear

7.10 Document Template Model

Fields:

- id
- businessId
- documentType
- name
- designVariant
- pageSize
- isDefault
- config (JSONB)
- isActive
- createdAt
- updatedAt

7.11 Audit Log Model

Fields:

- id
- userId
- documentId (nullable)
- action
- details (JSONB)
- createdAt

---

8. ENUM DEFINITIONS

DocumentType:

- INVOICE
- DELIVERY_CHALLAN
- CASH_MEMO
- MONEY_RECEIPT
- QUOTATION
- PURCHASE_ORDER
- WORK_ORDER
- OTHER

DocumentStatus:

- DRAFT
- ISSUED
- CANCELLED

PartyType:

- CUSTOMER
- SUPPLIER
- CONTRACTOR
- BOTH
- OTHER

ProjectType:

- TENDER
- LOCAL

ProjectStatus:

- PLANNING
- ACTIVE
- COMPLETED
- CANCELLED

UserRole:

- OWNER
- ADMIN
- OPERATOR
- VIEWER

DesignVariant:

- ORIGINAL
- MODERN

DiscountType:

- FIXED
- PERCENTAGE

PartyRole:

- CUSTOMER
- SUPPLIER
- CONTRACTOR
- RECEIVER
- OTHER

---

9. DOCUMENT NUMBERING SYSTEM

Every business and document type must have independent numbering.

Example:

ATI:

- ATI-INV-0001
- ATI-DC-0001
- ATI-CM-0001
- ATI-MR-0001

ARJ Tex:

- ARJ-INV-0001
- ARJ-DC-0001
- ARJ-CM-0001

Requirements:

- Configurable prefix.
- Configurable starting number.
- Configurable padding.
- Business-specific sequence.
- Document-type-specific sequence.
- Optional fiscal-year-based numbering.

- Atomic sequence generation using database transactions.
- Unique database constraint.
- Generate final document number when issuing, not while saving a draft.

Numbering must remain safe under concurrent document creation.

---

10. DOCUMENT CALCULATION LOGIC

For financial documents:

Item Amount = Quantity × Rate

Subtotal = Sum of Item Amounts

Discount:

- Fixed amount
- Percentage

Tax Amount = Applicable Tax Calculation

Grand Total = Subtotal - Discount + Tax

Requirements:

- Use Prisma Decimal / PostgreSQL DECIMAL for financial values.
- Do not rely on JavaScript floating-point calculations.
- Perform authoritative calculations on the server.
- Validate quantity, rate, discount, and tax values.
- Generate amount in English words.
- Support BDT as the default currency.
- Make financial fields optional for non-financial documents.

Delivery Challan must support quantity-only items without Rate and Amount.

Money Receipt must support an amount and payment information without requiring document items.

---

11. DOCUMENT-SPECIFIC FIELDS

Invoice

- Customer
- Invoice Number
- Date
- Reference
- Item Description
- Quantity
- Unit
- Rate
- Amount
- Discount
- Tax
- Total
- Amount in Words
- Signatures

Delivery Challan

- Customer
- Challan Number
- Date
- Vehicle Number
- Driver Name
- Item Description
- Quantity
- Unit
- Remarks
- Total Quantity
- Receiver Signature
- Authorized Signature

Cash Memo

- Customer
- Date
- Item Description
- Quantity
- Rate
- Amount
- Total
- Amount in Words
- Seller Signature

Money Receipt

- Receipt Number
- Date
- Received From
- Amount
- Amount in Words
- Payment Method
- Payment Reference
- Purpose
- Received By
- Authorized Signature

Quotation

- Quotation Number
- Customer
- Date
- Valid Until
- Subject
- Items
- Price
- Discount
- Tax
- Terms and Conditions
- Delivery Terms

Purchase Order

- PO Number
- Supplier
- Date
- Expected Delivery Date
- Items
- Quantity
- Rate
- Amount
- Payment Terms
- Delivery Address
- Authorized Signature

Work Order

- Work Order Number
- Issued To
- Date
- Project

- Work Description
- Location
- Deadline
- Terms and Conditions
- Authorized Signature

Other

- Configurable document title.
- Configurable custom fields.
- Custom item table.
- Custom terms and notes.
- Selectable print template.

Document-specific fields should be configurable. Use structured columns for frequently queried data and JSONB for additional configurable fields.

---

12. PRINT & PDF GENERATION

Printing accuracy is a core requirement.

Use:

- React-based document templates.
- HTML and CSS.
- Dedicated print routes.
- CSS @media print.
- CSS @page.
- Playwright for server-side PDF generation.
- Noto Sans Bengali for Bangla text.
- Appropriate English fonts.
- Proper page-break handling.

Two print variants:

A. Original Design

- Recreate existing printed forms.
- Preserve logo, header, watermark, table positions, dotted lines, and signature positions.
- Support custom paper dimensions.
- Support multiple copies per page.
- Match the uploaded samples as closely as possible.

B. Modern Design

- Professional layout.
- Clean header.
- Structured item table.
- Proper spacing.
- Clear totals.
- Signature section.
- Business branding.

Each business and document type may have different templates.

Original template dimensions must be established by inspecting the actual physical forms or their scans.

Printing must be tested with both browser printing and generated PDFs.

---

13. API ARCHITECTURE

Base URL: /api/v1

Business:

- GET /businesses
- POST /businesses
- GET /businesses/
- PATCH /businesses/

Parties:

- GET /parties
- POST /parties
- GET /parties/
- PATCH /parties/
- DELETE /parties/

Projects:

- GET /projects
- POST /projects
- GET /projects/
- PATCH /projects/

Products:

- GET /products
- POST /products
- GET /products/
- PATCH /products/

Documents:

- GET /documents
- POST /documents
- GET /documents/
- PATCH /documents/
- POST /documents//issue
- POST /documents//cancel
- POST /documents//duplicate
- GET /documents//print
- GET /documents//pdf

Templates:

- GET /templates
- POST /templates
- PATCH /templates/
- DELETE /templates/

Reports:

- GET /reports/summary
- GET /reports/documents
- GET /reports/projects

All APIs must enforce authentication, authorization, validation, and business-level data isolation.

---

14. NEXT.JS PROJECT STRUCTURE

business-document-system/

```
├── prisma/
│   ├── schema.prisma
│   ├── migrations/
│   └── seed.ts
```

```
│
├── public/
│   ├── fonts/
│   ├── logos/
│   └── images/
│
├── src/
│   ├── app/
│   │   ├── (auth)/
│   │   │   └── login/
│   │   ├── (dashboard)/
│   │   │   ├── dashboard/
│   │   │   ├── businesses/
│   │   │   ├── parties/
│   │   │   ├── projects/
│   │   │   ├── products/
│   │   │   ├── documents/
│   │   │   ├── templates/
│   │   │   ├── reports/
│   │   │   └── settings/
│   │   ├── print/
│   │   │   └── documents/[id]/
│   │   └── api/v1/
│   │
│   ├── components/
│   │   ├── ui/
│   │   ├── layout/
│   │   ├── forms/
│   │   └── shared/
│   │
│   ├── features/
│   │   ├── businesses/
│   │   ├── parties/
│   │   ├── projects/
│   │   ├── products/
│   │   ├── documents/
│   │   ├── templates/
│   │   └── reports/
│   │
│   ├── services/
│   │   ├── business.service.ts
│   │   ├── document.service.ts
│   │   ├── sequence.service.ts
│   │   ├── calculation.service.ts
```

```
│   │   └── pdf.service.ts
│   │
│   ├── lib/
│   │   ├── prisma.ts
│   │   ├── auth.ts
│   │   ├── permissions.ts
│   │   ├── validations/
│   │   ├── calculations/
│   │   └── utils/
│   │
│   ├── templates/
│   │   ├── original/
│   │   └── modern/
│   │
│   └── types/
│
├── .env
├── package.json
├── next.config.ts
└── tsconfig.json
```

Use feature-based modular architecture. Keep UI components, validation schemas, and business logic separated.

---

15. SECURITY & DATA INTEGRITY

Mandatory requirements:

1. Authenticate every protected request.
2. Validate user access to the selected business.
3. Apply businessId filtering to every business-specific query.
4. Never trust businessId supplied by the client without authorization checks.
5. Use server-side validation with Zod.
6. Use Prisma Decimal for financial calculations.
7. Preserve snapshots of business, party, and document item information.
8. Use database transactions for issuing documents.
9. Prevent duplicate document numbers.
10. Do not allow direct deletion of issued documents.
11. Record important actions in audit_logs.
12. Restrict sensitive settings to authorized roles.
13. Use secure password hashing.

14. Protect uploaded files and validate file types.
15. Implement regular PostgreSQL backups.
16. Do not expose internal IDs or data across unauthorized business boundaries.

For a multi-business setup, Business Membership and authorization must be checked on the server, not only in the frontend.

---

16. DEVELOPMENT PHASES

Phase 01 — Project Initialization

- Create Next.js project.
- Configure TypeScript.
- Install Tailwind CSS and shadcn/ui.
- Configure PostgreSQL.
- Install Prisma.
- Configure environment variables.

Phase 02 — Database Architecture

- Implement Prisma Schema.
- Validate relationships.
- Run initial migration.
- Seed ATI and ARJ Tex business profiles.
- Create initial owner account.

Phase 03 — Authentication & Dashboard

- Login system.
- Authentication.
- Role management.
- Dashboard layout.
- Business switcher.

Phase 04 — Business Management

- Business CRUD.
- Party CRUD.
- Project CRUD.
- Product CRUD.

Phase 05 — Document Engine

- Unified document CRUD.
- Dynamic item management.
- Document type configuration.
- Calculation engine.
- Automatic document numbering.
- Draft, Issue, Cancel workflow.

Phase 06 — Print Template Engine

- Original invoice.
- Modern invoice.
- Original delivery challan.
- Modern delivery challan.
- Cash memo.
- Remaining document templates.
- Print preview.
- PDF generation.

Phase 07 — Reports & History

- Document search.
- Filters.
- Project reports.
- Business reports.
- Audit history.

Phase 08 — Testing & Deployment

- Functional testing.
- Calculation testing.
- Numbering concurrency testing.
- Print alignment testing.
- Business isolation testing.
- Production deployment.
- Backup configuration.

Estimated MVP development duration: 9–13 working days, subject to template complexity and client feedback.

---

17. MVP SCOPE

Must implement:

- Multi-business management.
- Authentication.
- Business switcher.
- Client and supplier management.
- Tender and local project management.
- Product catalog.
- Eight document types.
- Unified document generation engine.
- Automatic document numbering.
- Draft / Issue / Cancel workflow.
- Original and Modern templates.
- Print and PDF.
- Document search and filtering.
- Basic reporting.
- Audit history.

Not included in initial MVP:

- Full accounting system.
- Inventory and stock management.
- Advanced customer ledger.
- Automatic payment reconciliation.
- Payroll.
- Mobile application.
- Advanced financial analytics.

These can be added in future versions.

---

18. IMPLEMENTATION PRIORITIES

Follow this development order:

1. Project setup and database.
2. Authentication and Business Switcher.
3. Business, Party, Project, and Product Management.
4. Unified Document Engine.
5. Document Numbering.
6. Invoice Template.
7. Delivery Challan Template.
8. Cash Memo Template.
9. Remaining Document Templates.

10. Print and PDF.
11. Reports.
12. Testing and Deployment.

Do not start by building all document templates independently. First build and test the unified document engine.

---

19. ACCEPTANCE CRITERIA

The MVP will be considered ready when:

- The owner can manage both businesses from one account.
- Each business has independent settings and document numbering.
- Customers and suppliers can be created and reused.
- Tender and local projects can be created.
- All eight document categories can be generated.
- Custom product descriptions can be entered.
- Document totals are calculated correctly.
- Challans can be created without financial fields.
- Draft documents can be edited.
- Issued documents cannot be silently modified or deleted.
- Document numbers remain unique under concurrent requests.
- Original and Modern print formats work.
- PDFs display Bengali and English text correctly.
- Existing documents retain their historical business and item information.
- Users cannot access another business's unauthorized records.
- Documents can be searched and filtered.
- Important actions are recorded in audit history.

---

20. IMPORTANT DEVELOPMENT INSTRUCTIONS FOR AI CODING AGENTS

When implementing this project, follow these rules:

1. Read this specification completely before writing code.
2. Do not change the agreed architecture without explaining the reason.
3. Use TypeScript throughout the application.

4. Use PostgreSQL with Prisma ORM.
5. Follow modular, reusable, and maintainable coding practices.
6. Do not create separate backend CRUD systems for each document type.
7. Build one reusable Document Engine.
8. Keep financial calculations on the server.
9. Use database transactions for document issuance and numbering.
10. Preserve historical document snapshots.
11. Enforce business-level authorization on every protected operation.
12. Use reusable React components.
13. Keep document print templates independent of dashboard layouts.
14. Follow responsive design principles.
15. Validate every form and API request.
16. Write appropriate error handling.
17. Avoid unnecessary dependencies.
18. Do not implement out-of-scope ERP features.
19. Complete and test one development phase before moving to the next.
20. Maintain documentation and update the implementation checklist.

Development Approach:
Work phase by phase. Before starting each phase, identify its tasks and dependencies. After implementation, verify the functionality, report completed work, list remaining issues, and then proceed to the next phase.

END OF PROJECT SPECIFICATION